

Persistence Mechanics: Architecture, Dynamics, and the Operational Action

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Before a complex system can adapt or self-organize, it must remain identifiable through time. From biological organisms to computational processes, disparate systems share a single existential mandate: maintaining structural coherence against entropic erasure. We formalize the mechanics of this persistence, deriving the substrate-independent, minimal architecture. We show that any persistent system must satisfy three fundamental constraints: Finiteness (bounded resources), Conservation (preserved information), and Causality (well-defined evolution), yielding six irreducible pillars: Capacity, Map, Protocol, Governor, Toll, and Margin. This static architecture is grounded in time by defining the minimal dynamical description of state transitions: toll (c), structural complexity (K), and coordinated transition flux (f) whose product defines the dissipation density $\mathcal{D} = c \cdot K \cdot f$. Integrating $\mathcal{D}$ over a system's history defines the Operational Action S_{Φ} . The resulting Selection Principle shows that minimum-action configurations asymptotically dominate observable histories. Mapping systems onto this architectural invariant provides a common basis for comparing maintenance constraints, identifying structurally analogous failure modes, and formulating cross-domain hypotheses for subsequent validation.

I. INTRODUCTION

From biological organisms to engineered feedback loops, a central problem in complex-systems science is how systems retain recognizable organization despite disturbance, turnover, and decay [1]. Several mature disciplines address particular problems faced by such systems: control theory characterizes regulation and stability; information theory characterizes representation and communication under noise; and thermodynamics constrains energy conversion and dissipation. Each typically begins with a system boundary, state space, and relevant mechanisms already specified. This leaves a prior cross-domain question: what minimal architecture must any system instantiate simply to remain identifiable through time?

Previous complex-systems research has explored complementary aspects of persistence. Prigogine characterized minimum entropy production in stationary states under linear near-equilibrium conditions [2]; Friston related adaptive self-maintenance to variational free-energy minimization through an internal model [3]; England examined how external driving can favor particular configurations in nonequilibrium systems [4]; and cybernetics established organizational constraints on effective regulation, including requisite variety and the Good-Regulator theorem [5, 6]. These approaches characterize complementary aspects of persistence. Comparing them across domains requires a common criterion for what it means for a system to persist.

We say that a system *persists* over an observation interval when, relative to a specified system boundary and class of disturbances, it retains the organization by which it is identified and remains within its viable operating conditions. Persistence does not require an unchanging

microscopic state: components may turn over, resources may flow, and internal states may change, provided the system's defining organization remains intact.

Persistence Mechanics addresses three questions: what must be instantiated for persistence, how the resulting architecture operates through time, and how its cumulative costs affect which histories remain observable.

Section II derives the **Architecture of Persistence**: six substrate-independent pillars derived from the logical requirements of Finiteness, Conservation, and Causality. Section III develops the **Dynamics of Persistence**, identifying the transition toll c , structural complexity K , and coordinated transition flux f . Section IV derives the exponential weighting of histories by cumulative Operational Action and states the conditions under which surviving histories concentrate on minimum-action trajectories. Section V extends the additive baseline to systems with external flux and shared-resource contention. Section VI illustrates how the six pillars are instantiated across biological, ecological, and computational substrates.

II. THE ARCHITECTURE OF PERSISTENCE

Before a system can regulate, compute, or evolve, it must remain within the conditions that preserve its identity. We organize those conditions along three dimensions: bounded resources, retention of organization, and ordered evolution. Distinguishing the mechanisms that establish these conditions from those that protect them motivates six functional pillars. This section defines those pillars, examines the failure modes they address, and describes how familiar maintenance capabilities can be composed from them.

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A. The Structural Constraints

Any process persisting through time is bounded in extent, carries distinguishable content, and evolves through an ordered sequence of states. Absent any one, it is indistinguishable from its environment, from noise, or from something static. Formally, these correspond to: **Finiteness** (bounded resources against divergence), **Conservation** (preserved information against dissipation), and **Causality** (well-defined temporal evolution against ambiguity).

Existence, however, is transient. To persist, each constraint must survive its characteristic perturbation. Each therefore demands **provision** (establishing the means to satisfy the constraint) and **defense** (protecting that means from the characteristic failure mode) yielding the six irreducible pillars.

Constraint	Provision	Defense
Finiteness	Capacity	Margin
Conservation	Map	Protocol
Causality	Toll	Governor

Table I. The three constraints, organized by provision and defense, define the six pillars of the PM architecture.

A *pillar* is a functionally distinct condition that must be materially or informationally instantiated for persistence. Pillars need not correspond one-to-one with components: a single mechanism may instantiate several pillars, and one pillar may be distributed across many mechanisms.

These pillars operate upon a medium that imposes immutable limits (bandwidth, latency, noise floor) referred to as the **Substrate**.

B. The Six Pillars

The following provide concrete realizations of the six pillars showing that each pillar has a convergent theorem from six unrelated fields.

Capacity (CAP): The infrastructure required to acquire resources, process signals, and perform work. It defines the system’s operational potential. *Convergent Evidence (The Bekenstein Bound [7]):* Bekenstein’s bound limits the information content of any finite region, and the Margolus-Levitin limit[8] bounds the rate of state transitions; together they establish that finite capacity is a structural invariant.

Margin (MAR): The structural and energetic reserves held to absorb stochastic shocks. It defines the system’s ‘Safe Operating Area,’ providing the buffer required to survive environmental variance without terminal failure. (The gain and phase margin). *Convergent*

Evidence (Nyquist-Bode Stability Margins[9] and Kingman’s Formula[10]): A system operating at theoretical criticality has zero tolerance for variance. Kingman’s formula shows that as a queueing system’s utilization approaches unity, waiting time diverges asymptotically, zero buffer against arrival variance means unbounded delay. Nyquist’s stability margins are the linear-control manifestation of the same general requirement: persistent systems must maintain structural tolerance away from critical thresholds.

Map (MAP): The retained organization, relational structure, or state criterion that distinguishes the system’s viable configurations from its environment and from failed configurations. In adaptive systems, Map may be instantiated as an internal predictive model; in passive systems, it may be embodied directly in physical organization. *Convergent Evidence (The Good-Regulator Theorem [6]):* Conant and Ashby’s Good-Regulator theorem establishes that persistent regulation requires an internal model; without Map, the system–environment boundary dissolves.

Protocol (PRO): The communicative glue regulating information flow between pillars. It ensures internal coherence, standardizes interfaces, and minimizes signal decay across the system’s internal boundaries. *Convergent Evidence (Shannon’s Source-Channel Separation and Channel Capacity [11]):* Shannon’s coding results characterize conditions for reliable communication, providing a realization of the Protocol role without requiring a physically separate interface layer.

Toll (TOL): The finite temporal and resource requirements through which state transitions are physically or computationally realized. Toll need not be dissipative in every implementation; it records that ordered change must occur through a substrate with finite transition capabilities. *Convergent Evidence (Landauer’s Principle and Margolus-Levitin Limit [8, 12]):* On thermal substrates, irreversible operations (such as information erasure) carry a strict lower bound of $k_B T \ln 2$ (Landauer’s Principle), but even broadly, execution latency (Margolus-Levitin) makes Toll a structural requirement.

Governor (GOV): The active or passive constraint structure that keeps evolution within the system’s viable region by suppressing, redirecting, or terminating destabilizing transitions. *Convergent Evidence (Ashby’s Law of Requisite Variety [5]):* Ashby’s Law of Requisite Variety requires that the constraint mechanism match the variety of perturbations; without Governor, runaway is unavoidable.

With each pillar independently attested, the architecture becomes grounded with citable bounds and can be built on formally.

C. Necessity: Counterfactual Failure Modes

The necessity of the six pillars follows by analyzing the **Counterfactual Failure Mode** of a system lacking exactly one pillar. In every case, the system’s viability collapses, limiting its expected lifetime τ to the onset of its characteristic perturbation.

- **No Capacity (Starvation):** Without baseline operational infrastructure, the system cannot acquire, process, or export resources. Once existing reserves are spent, any perturbation is fatal; $\tau \rightarrow 0$ via rapid exhaustion.
- **No Margin (Fragility):** Operating at theoretical criticality leaves zero tolerance for environmental variance. Any non-zero perturbation exceeds the system’s structural limits immediately, so the expected time to a fatal shock is finite; $\tau \rightarrow 0$ by random shock.
- **No Map (Dissolution):** Without an internal model, the system has no reference for coherence, and no boundary to maintain against its environment; $\tau \rightarrow 0$ as the system–environment boundary dissolves.
- **No Protocol (Decoherence):** Components act on statistically independent, outdated states, with no mechanism to reconcile them; $\tau \rightarrow 0$ by organizational fragmentation.
- **No Toll (Zeno Collapse):** A causal sequence requires irreducibly discrete state transitions. Without an irreducible toll per operation, the sequence can be infinitely subdivided; no individual step remains distinguishable as a cause of the next, and the ordering required for well-defined evolution dissolves. $\tau \rightarrow 0$ by causal decoherence.
- **No Governor (Divergence):** Unconstrained positive feedback drives resource consumption toward unbounded growth, exhausting the substrate or diverging past any recoverable bound; $\tau \rightarrow 0$ by unbounded runaway.

Since the removal of any component results in $\tau \rightarrow 0$, every pillar is necessary.

D. Completeness: Compositional Closure

Proving sufficiency over an unbounded architectural space would require ruling out every possible construction, which no finite argument can do. Instead, every capability plausibly treated as an independent requirement has been found to decompose into the six pillars, with no counterexample identified to date.

Memory is Map content provisioned against temporal variance by Margin. Prediction is Map operation on incoming signal via Protocol. Repair is Governor-triggered

re-provisioning within the Margin reserve. Replication is Protocol-mediated write of Map onto fresh Capacity. Advanced topologies such as feedforward networks, state observers, and adaptive controllers extend the six-pillar structure the same way: none introduces a new primitive, each is a composition within the existing six.

E. The Cost of Persistence

A system pays continuously merely to remain distinguishable from its environment. The six pillars consume resources, acting as a continuous resistor against entropic erosion. This total structural cost provides the scalar metric required to compare systems.

Every pillar imposes a strictly positive cost, partitioning into two operational roles:

- **Direct Structural Costs:** The overhead of maintaining the system’s baseline existence. Maintaining the vessel (Z_{CAP} , Z_{MAR}), the internal model (Z_{MAP}), and executing state updates (Z_{TOL}) requires a strict allocation of resources regardless of how the system is organized.
- **Coordination Costs:** Protocol (Z_{PRO}) and Governor (Z_{GOV}) incur direct structural costs to operate, but function as active constraints rather than passive structures. Their operational effect is to suppress uncoordinated variance. As in control theory, the active controller consumes continuous energy to operate, but its regulation ensures the closed loop’s total dissipation remains strictly lower than the open-loop alternative.

To quantify baseline structural costs independently of environmental work, the system is evaluated at its zero task-flux limit. This defines the *Quiescent Steady State* (QSS), a homeostatic ground state where internal house-keeping dissipation balances entropic decay. In this baseline, the six pillars operate along functionally uncoupled, orthogonal channels:

$$\bar{Z}_{net} = Z_{CAP} + Z_{MAP} + Z_{TOL} + Z_{MAR} + Z_{PRO} + Z_{GOV} \quad (1)$$

Because the pillars occupy orthogonal domains at zero flux, any correlation between them is already priced into the individual coordination terms Z_{PRO} and Z_{GOV} ; no additional cross-term is needed to capture coupling between pillars within this baseline. Coupling outside QSS introduces non-linear contention, which we formalize in section V on open systems.

F. Relation to Existing Disciplines

Existing fields studying systems that persist have encountered the six-pillar architecture already as they each

are functional instantiations. In each case, the discipline addressed operational problems that sit downstream of these six pillars, taking the essential architecture as given.

1. Relation to Information Theory

Information theory quantifies how representation, encoding, and noise constrain communication across physical mediums. In doing so, it relies on structural primitives that map directly to the six pillars: channel capacity (Capacity), codebook topologies (Map), framing and handshaking routines (Protocol), power constraints (Governor), processing overhead (Toll), and link margins (Margin). Information theory focuses on maximizing mutual information or minimizing error rates given a fixed architecture. Persistence Mechanics instead derives the architecture required for any state representation to survive noise and resist structural erasure.

2. Relation to Thermodynamics

Thermodynamics describes how physical systems evolve under energy exchange and thermal fluctuations. It relies on physical primitives that correspond to the six pillars: extensive state variables and phase space volume (Capacity), metastable energy landscapes (Map), kinetic transition pathways (Protocol), conservation laws (Governor), irreversible entropy production (Toll), and free-energy activation barriers (Margin). Classical thermodynamics assumes the pillars, an external heat bath and continuous state variables, as given boundary conditions. Where thermodynamics describes how heat flows, persistence mechanics specifies what structural architecture must exist before heat, flow, or temperature can be meaningfully defined.

3. Relation to Control Theory

Classical control theory emerged from engineering physical servomechanisms: radar mounts, autopilots, and chemical plants whose historical context was optimizing systems already built, powered, and staffed. Its four given components (the plant, the sensor, the controller, and the actuator) are convergent evidence for four of the six pillars: Plant/Actuator (Capacity), Reference Signal/Setpoint (Map), Sensor/Feedback Path (Protocol), Controller/Limiter (Governor). The remaining two pillars are rarely canonized alongside the first four as they are not physical components, but describe how the four components behave under time and stress. Time delays, bandwidth limits, and processing latency (Toll) were minimized as engineering overhead and gain/phase margins (Margin) were tuned for stability. Control theory

asks how to regulate an existing system and the classical four-component loop is a physical instantiation of the six-pillar invariant.

III. THE DYNAMICS OF PERSISTENCE

To describe how a system maintains its architecture through time, we first identify the operational requirements of its state transitions: the work performed per update, the cost of that work, and the frequency of updates. These three quantities define a local operational cost rate. We then examine the constraints imposed by a finite resource budget to measure the cumulative cost of a history.

A. State Transition Costs

The operation of the six pillars through time introduces the constraint that transitions must be realized within the substrate's available resources and response times. Quantifying this requires distinguishing the work an update requires, the cost of that work, and the update frequency.

1. **Structural workload (K):** The number of elementary operations required for one maintenance update of the local structure. An update preserves, repairs, or modifies the organization through which the pillars are instantiated. Here, K measures operations per update. Its relationship to algorithmic description length [13] is implementation-specific. *Domain Manifestation (Computational Workload):* In a computational implementation, K counts the elementary operations executed by an update procedure under a specified computational model.
2. **Operational toll (c):** The mean resource cost per elementary operation, expressed in a specified accounting unit. Substrate manifestations include dissipated energy, computational expenditure, or other explicitly defined resources. Transition latency imposes a separate constraint on achievable update rates; it is not automatically equivalent to dissipative cost. *Domain Manifestation (Information Erasure):* In the standard isothermal setting, erasing one unbiased bit requires at least $k_B T \ln 2$ of heat dissipation [12]. This provides a physical lower bound on the toll for that class of operation.
3. **Coordinated transition flux (f):** The number of maintenance updates performed per unit substrate measure per unit time. An update involves coordinated operations, whose total is recorded by K . The flux required for persistence is dictated by the system's organization and the environmental disturbances it must accommodate. *Domain*

Manifestation (Communication-Constrained Control): Data-rate theorems establish communication requirements for stabilizing specified dynamical systems [14]. For updates carrying a fixed amount of transmitted information, such requirements can constrain update frequency.

While a system might be made up of sub systems each performing transitions, the parent system transition costs will only be the sum of the subsystem costs.

B. Operational Cost Rate Density

Repeated operations draw on a finite resource budget. Comparing this demand with available resources requires combining the workload, operational toll, and update flux into a local cost rate per unit substrate measure.

$$\begin{aligned} [c] &= \frac{\text{cost}}{\text{operation}} \\ [K] &= \frac{\text{operations}}{\text{update}} \\ [f] &= \frac{\text{updates}}{\text{substrate measure} \cdot \text{time}} \end{aligned}$$

For transitions requiring K elementary operations each, if the mean cost of those operations is c , the cost per update is Kc . Performing updates at flux f therefore yields the local density:

$$\mathcal{D} = cKf, \quad [\mathcal{D}] = \frac{\text{cost}}{\text{substrate measure} \cdot \text{time}} \quad (2)$$

The interpretation of $\mathcal{D}$ depends on the resource used to define c . When c measures dissipated energy per operation, $\mathcal{D}$ is a dissipation density; otherwise it records the corresponding operational resource cost.

When any operational cost is 0, $\mathcal{D} = 0$. Whether the corresponding inactivity or absence of expenditure compromises persistence depends on the maintenance required over the observation interval.

C. The Complexity Trade-off

Transitions draws on a resource budget whose sustainable rate and ceiling $\mathcal{D}_{\max}$ is set by the substrate. This results in a competition among the three factors.

Transitions draw on a resource budget whose sustainable rate $\mathcal{D}_{\max}$ is set by the substrate and environment. This budget constrains the combinations of workload, toll, and transition flux that the system can sustain.

Proposition 1 (The Complexity Trade-off). *For homogeneous operation activity in a fixed substrate region, operation within the sustainable budget requires*

$$cKf \leq \mathcal{D}_{\max}. \quad (3)$$

For $c, K > 0$, the corresponding transition-flux bound is

$$f \leq \frac{\mathcal{D}_{\max}}{cK}. \quad (4)$$

At a fixed, binding budget, increasing workload requires a compensating reduction in toll, transition flux, or both.

Proof. Substitute $\mathcal{D} = cKf$ into the budget constraint and divide by $cK > 0$. $\square$

For illustration, consider a cell with a fixed ATP budget. If each update requires twice as many ATP-consuming operations while the ATP cost per operation remains unchanged, the same budget supports only half as many updates per unit time. Maintaining the original update rate would require either a lower cost per operation or a larger budget.

D. Operational Action

While the operational cost rate describes instantaneous expenditure, evaluating the accumulated burden of distinct histories requires integrating this rate across the system's spatial extent and temporal duration.

Let $\Sigma(t)$ denote the physical or computational elements defining the system boundary at time t . Integrating the local cost-rate density over this footprint yields the total instantaneous operational cost:

$$Z_{\text{net}}(t) = \int_{\Sigma(t)} \mathcal{D}(x, t) d\mu(x). \quad (5)$$

Here, μ specifies the substrate measure used to define $\mathcal{D}$. For discrete elements, the counting measure reduces this integral to a sum over the localized elements.

The architectural pillars need not occupy physically disjoint regions of the substrate. Because the six pillars represent distinct functional requirements, their total expenditure partitions exactly into their additive costs (section II E), yielding $Z_{\text{net}}(t) = \sum_i Z_i(t)$. This strict accounting ensures that even if a single physical mechanism multiplexes multiple pillars, its operational cost is uniquely apportioned without double-counting.

The *Operational Action* S_{Φ} of a structural history Γ over an observation interval $[0, T]$ is defined as its cumulative operational cost:

$$\begin{aligned} S_{\Phi}[\Gamma; T] &= \int_0^T Z_{\text{net}}(t; \Gamma) dt \\ &= \int_0^T \int_{\Sigma(t)} \mathcal{D}(x, t; \Gamma) d\mu(x) dt. \end{aligned} \quad (6)$$

Comparing competing systems requires evaluating this action over a common observation interval, a shared resource unit, and a consistent convention for identifying the system boundary.

The dimensions of S_Φ inherit the units of the operational toll c . If c represents dissipated energy per operation, S_Φ quantifies macroscopic dissipated energy. If c is defined as a dimensionless action per operation, S_Φ becomes a pure, dimensionless count of state transitions. In arbitrary physical systems, a dimensionless Operational Action can always be recovered by dividing S_Φ by a characteristic reference cost.

While the static architecture identifies the functions instantiated by the system, the Operational Action records the exact thermodynamic burden of their realized operation. This cost accumulates along any history, whether or not the system ultimately persists. Relating this accumulated Operational Action to the probability of survival requires a specific failure model, formalized in section IV.

IV. MINIMUM ACTION AS SURVIVORSHIP

The Operational Action defines what a system must minimize, but not which histories survive to be observed. The selection filter derives from the multiplicative compounding of independent transition risk, and show that only minimal-action configurations remain observable.

A. The Exponential Filter

If persistence is maintained by state transitions, and each transition risks scrambling information into the substrate noise floor, then a history of N transitions survives only if every step survives. Independent errors compound multiplicatively, yielding an exponential filter.

Consider a structural history Γ accumulating operational action $S_\Phi[\Gamma]$ (equation (6)). Because the substrate possesses a finite noise floor, each elementary transition carries an intrinsic risk of information loss. Let q be the probability that a single elementary transition maintains causal coherence.

In QSS, successive transitions are statistically independent dissipative events. The probability that Γ maintains coherence is therefore multiplicative:

$$P(\text{survival} \mid \Gamma) = q^{S_\Phi[\Gamma]} \quad (7)$$

Using the identity $q^C = e^{C \ln q}$ with $C = S_\Phi[\Gamma]$, and defining $\beta \equiv -\ln q > 0$ (positive since $0 < q < 1$), we have $S_\Phi[\Gamma] \ln q = -\beta S_\Phi[\Gamma]$, yielding the survival probability distribution:

$$P(\text{survival} \mid \Gamma) = e^{-\beta S_\Phi[\Gamma]} \quad (8)$$

Here $\beta \equiv -\ln q$ is the substrate's inverse operational tolerance, a geometric invariant of the channel. For discrete substrates, q is determined by the channel geometry; in continuous media, β emerges from the substrate's noise spectral density.

This is the informational analogue of the Boltzmann distribution $P \propto e^{-E/k_B T}$, with S_Φ as energy and β as inverse temperature. The Boltzmann form requires ensemble equilibrium and a thermal bath; the present form requires only independent transition risk along a single history and a substrate noise floor. The informational route recovers the same exponential law under strictly weaker assumptions: substrate geometry replaces thermodynamic equilibrium.

B. The Severity of the Exponential Filter

The operational severity of this filter can be quantified by comparing trajectories. Let $\Gamma_{\min}$ denote the minimal-action trajectory and Γ_{var} any non-minimal path with an action deviation $\delta S_\Phi > 0$. Over any observation window, the relative survival probability is:

$$\frac{P(\Gamma_{\text{var}})}{P(\Gamma_{\min})} = e^{-\beta \delta S_\Phi} \quad (9)$$

When the fundamental update cycle is much shorter than the observation timescale, even a small per-transition deviation compounds into a vast accumulated δS_Φ , and the survival ratio $P(\Gamma_{\text{var}})/P(\Gamma_{\min})$ falls to zero. The gap between the best trajectory and the next-best widens without bound.

Because survival probability depends exponentially on S_Φ , only histories approaching the minimum retain non-negligible observation probability:

The Selection Principle: Among all configurations capable of encoding a given structural complexity, only those possessing the minimal possible Operational Action (S_Φ) asymptotically dominate the observable distribution.

Because S_Φ counts elementary transitions, it is the cumulative ledger of a system's dissipation history. The substrate does not optimize this ledger; it erases ledgers that fail to balance.

C. Eliminative vs. Teleological Selection

The severity of the exponential filter is a statement about the losers, not the winner. It says nothing about whether $\Gamma_{\min}$ is efficient, elegant, or well-suited to anything, only that every alternative to it decayed faster. A sub-optimal trajectory survives just as certainly as an optimal one, provided everything else was worse. What looks like design, foresight, or purpose is not a property of the winner; it is the absence of its competition.

D. Minimum Action as Survivorship

This eliminative mechanism is formalized as:

Proposition 2 (Minimum Action as Survivorship). *In QSS, the observable trajectories of any persistent system are those achieving the minimal Operational Action S_Φ among all candidate trajectories.*

Proof. By equation (8), the survival probability of any history Γ is $P(\text{survival} \mid \Gamma) = e^{-\beta S_\Phi[\Gamma]}$. For any non-minimal trajectory Γ_{var} with $\delta S_\Phi = S_\Phi[\Gamma_{\text{var}}] - S_\Phi[\Gamma_{\text{min}}] > 0$, equation (9) gives $P(\Gamma_{\text{var}})/P(\Gamma_{\text{min}}) = e^{-\beta \delta S_\Phi}$. As δS_Φ accumulates over the observation window, this ratio vanishes. Therefore, only trajectories achieving the minimal Operational Action retain non-negligible observation probability. The observable set is the argmin of $S_\Phi[\Gamma]$. $\square$

Standard quantum mechanics grounds the Principle of Least Action in phase interference. The Selection Principle offers a distinct, classical mechanism (informational elimination) by which candidate physical histories are selected: the substrate itself acts as a filter, and what remains observable is only what has not yet dissipated.

V. OPEN SYSTEMS: NECESSITY WITHOUT ADDITIVITY

The Selection Principle and the Principle of Least Action were derived in the Quiescent Steady State: the zero-flux limit where no net energy or information enters or leaves. Many systems, however, exchange flux with their environment to persist. We show that the necessity of the six pillars survives this extension unchanged; only the net impedance (Z_{net} , section II E) requires re-derivation in flux-driven capacity.

A. Pointwise Necessity and Operational Latency

Because the substrate continuously filters for causal coherence, survival is an instantaneous requirement, not a time-averaged one.

Proposition 3 (Pointwise Necessity). *For any system persisting past instant t , all six pillars must be structurally instantiated at t , regardless of external flux.*

Proof. By the counterfactual failure modes of section II C, the absence of any pillar drives the corresponding primitive to zero ($c \rightarrow 0$, $K \rightarrow 0$, or $f \rightarrow 0$), leaving the system entirely vulnerable to its corresponding failure mode. Under continuous environmental perturbation, the instantaneous hazard rate diverges, driving the probability of surviving any finite interval dt to zero. The requirement is instantaneous and flux-independent. $\square$

Flux alters a pillar's *operation*, not its *existence*. During phases of rapid growth or unbounded resource influx, a control mechanism may exhibit low activation ($f_{\text{Gov}} \approx 0$), minimizing active coordination expenditure. Yet its *structural capacity* must exist at t ; if the mechanism is fundamentally absent, the system fails to persist the moment it reaches environmental saturation.

B. Open Systems and the Dynamic Budget

Strict additivity ($\bar{Z}_{\text{net}} = \sum_i Z_i$) relies on the Quiescent Steady State, where orthogonal domains prevent interference. In open systems, external flux raises the dissipation ceiling; as operations scale against this moving ceiling, resource contention forces non-linear cross-coupling $\Delta(t)$ between previously orthogonal pillars:

$$Z_{\text{net}}(t) = \sum_i Z_i(t) + \Delta(t). \quad (10)$$

A stochastic derivation of $\Delta(t)$ from substrate-native properties is developed in appendix A; three regimes emerge as its limits, where ρ is the dimensionless coordination load:

1. **Linear-Response Regime (Low ρ):** $\Delta \rightarrow 0$; QSS additivity is the leading-order behavior.
2. **Saturation-Divergence Regime (High ρ):** Coordination cost diverges as the dissipation ceiling is approached, forcing structural pauses (apoptosis, circuit breakers) to shed load.
3. **Correlated-Demand Regime (Non-Markovian):** Relaxing the memoryless assumption recovers the standard heavy-traffic generalization, with the $\rho/(1 - \rho)$ form as its memoryless special case.

The derivation does not alter the necessity claim. The six pillars remain structurally mandatory regardless of the specific form of $\Delta(t)$; the idealized limit where $\Delta(t) \rightarrow 0$ recovers QSS additivity.

That derivation also provides a unified account of these three regimes, which were previously analyzed independently. Linear-response recovery, heavy-traffic divergence, and the correlated-demand generalization were each discovered independently in unrelated fields. Here they emerge as three limits of one substrate function.

VI. UNIVERSALITY MAPPING: ARCHITECTURE AND DYNAMICS

The core mechanics of persistence are identifiable across systems of any substrate. Mapping both the static

architecture (the six pillars) and the operational dynamics (the dissipation trade-off and selection filter) translates domain-specific phenomena into identical mechanical constraints. The following mappings demonstrate architectural universality across distinct physical and informational domains. Detailed operationalizations belong to applied persistence mechanics; here, we illustrate the underlying structural invariants.

A. Biological Cell

Formalizing the biological concept of autopoiesis [15]:

Architecture: Capacity as metabolic hardware; Map as genome; Protocol as genetic code and signaling machinery; Governor as apoptosis and cell-cycle checkpoints; Toll as the thermodynamic cost of proofreading; Margin as ATP reserves and stress buffers.

Dynamics: The cell’s basal metabolic rate ($\mathcal{D}$) is the product of its ATP cost per molecular reaction (c), proteome complexity (K), and metabolic turnover rate (f). An organism cannot simultaneously maximize genomic complexity and reproductive speed without exceeding the local thermodynamic ceiling. Trajectories that overdraw this budget undergo exponential suppression via starvation or competitive exclusion.

Cross-Domain Transfer: Rather than treating aging as a correlational pattern of stochastic damage, this architectural defines senescence as the progressive degradation of operational pillars across nested sub-systems, revealing structural failure modes analogous to resource exhaustion in computational systems.

B. Ecological Network

Formalizing ecological ascendancy and network thermodynamics [16]:

Architecture: Capacity as biomass and solar energy capture; Map as trophic structure and niche topology; Protocol as nutrient cycling and metabolic exchange; Governor as carrying-capacity and predator-prey feedbacks; Toll as thermodynamic heat loss per trophic transfer; Margin as biodiversity and structural redundancy.

Dynamics: Total ecosystem respiration ($\mathcal{D}$) is the product of heat lost per transfer (c), food-web complexity (K), and the rate of biomass turnover (f). Because solar influx rigidly caps $\mathcal{D}$, overly complex food webs with inefficient trophic transfers are systematically pruned by famine and ecological collapse, leaving only the least-action trophic topology.

Cross-Domain Transfer: Mapping ecological networks to energy grids and financial markets reveals that cascading collapse obeys the same fixed dissipation

ceiling, making grid load-shedding and market circuit-breakers candidate tools for anticipating tipping points in biodiversity loss.

C. Large Language Model

Formalizing a young domain still establishing its foundations:

Architecture: Capacity as inference compute and GPU memory bandwidth; Map as parametric weights and attention matrices; Protocol as tokenization, layer connections, and key-value (KV) caching; Governor as output constraints and behavioral guardrails; Toll as autoregressive latency and state-update friction; Margin as unused context window, processing redundancy, and decision margin.

Dynamics: The model’s active dissipation burden ($\mathcal{D}$) is the product of compute friction per generated token (c), context and parameter footprint (K), and generation velocity (f). In autoregressive generation, attempts to maximize context length and reasoning depth without lowering parameter activation costs hit a hard memory bandwidth and compute ceiling. The Operational Action acts as a strict evolutionary filter on AI architectures: models that fail to minimize this product succumb to cumulative attention noise (hallucination) or catastrophic inference costs, driving the structural selection of minimal-action survivors (e.g., KV-cache distillation, sparse routing, or linear-time state-space models).

Cross-Domain Transfer: The comparatively young field of AI can draw on cross-disciplinary knowledge without re-deriving it: long-context degradation is currently benchmarked and treated empirically. If the Saturation-Divergence Regime correctly describes KV-cache saturation, its onset is a hazard-rate problem already solved by Weibull analysis.

Contention Regimes: The three regimes of appendix A are visible in inference. Short-context generation resembles the Linear-Response Regime. As context length approaches the memory-bandwidth ceiling, generation quality collapses in a manner resembling the Saturation-Divergence Regime, the KV-cache saturation and attention degradation already documented in long-context models. Non-uniform, correlated attention patterns (repetition, recency bias) degrade worse than a memoryless baseline would predict, suggesting the Correlated-Demand Regime as a candidate lens, not a demonstrated one.

D. Further Applications

The same six-pillar mapping extends to domains further removed from physical substrates. Two examples

illustrate distinct payoffs of that extension, briefly rather than in full: one shows the mapping makes previously incomparable systems commensurable; the other shows the architecture binds even a domain with no physical substrate at all.

- **Institutional Organizations:** A shared dissipation metric shows how bureaucratic decay in a growing firm and signal decay in a fiber-optic network are directly comparable structural failures.
- **Scientific Enterprise:** Reframing Planck’s observation that science advances one funeral at a time as the thermodynamic survival of funded believers over verification friction shows that theoretical paradigms are constrained by the same dissipation limits as cells or firms.

VII. CONCLUSION

Fluctuation theorems, master equations, and Langevin dynamics describe the statistics of transitions; the architecture of persistence describes the structure that must exist before any such statistics can be defined. For the fields of adaptation and self-organization, this shifts the fundamental question: before a system can adapt or self-organize, it must possess a minimal architecture permitted to exist.

We have developed that architecture and a mechanism of selection under the stated dynamical assumptions. The six pillars organize the requirements of persistence; the Operational Action measures cumulative dissipation; and the Selection Principle describes the resulting survival bias when failure hazard is proportional to that action.

The shared architecture provides a starting point for transferring knowledge across disciplines. Because this universal mapping projects every persistent system onto the same six-pillar substrate, it functions as a Rosetta stone across disciplines. Hard-won solutions in one domain become candidates for transfer to another: protocol engineering that prevents signal decay in a fiber-optic network can suggest mechanisms for preventing fragmentation in a corporate hierarchy. In emerging domains such as AI, where the pillars are present but unnamed, this vocabulary potentially lets systems be engineered for resilience without re-deriving structural knowledge from scratch.

The same mapping enables direct quantitative comparison among competing systems in a shared environment. By evaluating how competing organisms, firms, or tech-

nical architectures manage their finite dissipation budgets, relative operational resilience can be assessed prior to empirical failure.

Organizational decline, ecosystem collapse, and the limits of biological longevity have been described through domain-specific frameworks. By defining exactly what is required to persist, this mechanics recasts them as structural failures under a single, unified mechanics: disparate phenomena become comparable instances of the same dissipation constraint, rather than loose analogies.

The Complexity Trade-off is a pre-observational constraint on structural complexity, coordination speed, and energetic efficiency. At a binding dissipation ceiling, increasing one factor requires a compensating reduction in the product of the other two; given the ceiling, any two positive factors bound the third.

The most consequential limiting case is fundamental physics: whether standard physical law is itself a substrate-specific instantiation of this same invariant. Further exploration will determine whether this architectural boundary holds universally. Candidate mathematical languages for testing this include Information Geometry (continuous substrates), Arithmetic Geometry (discrete substrates), and Sheaf Theory (compositional substrates).

Natural avenues for the extension include systems composed of nested persistent subsystems, what it takes for a system to arise from disorder, and a fuller account of decay itself: this paper identifies failure modes, without characterizing the trajectory a system follows on the way there, or whether decay has its own structure worth deriving.

Persistence Mechanics does not explain why structure arises. It explains why only certain architectures remain to be observed. Persistence requires no intent, no optimization, and no forward-looking agency, only that the substrate continuously filter non-viable configurations. What remains observable is the minimal architecture of persistence.

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Appendix A: An Example of the Dynamic Budget via Queueing-Theoretic Contention

Open systems violate the orthogonality that made $\bar{Z}_{\text{net}}$ strictly additive in QSS. To quantify the resulting contention between Protocol and Governor, we derive the cross-coupling $\Delta(t)$ in equation (10) from discrete substrate properties.

1. Coordination capacity and load

Protocol and Governor share the substrate that performs all state transitions. The Margolus-Levitin limit bounds the minimum time between distinguishable operations, yielding a substrate-level coordination rate f_{max} . In QSS, the two pillars operate on orthogonal domains (their demands share no microstate), so their loads add.

With $f_{\text{PRO}} + f_{\text{GOV}} \ll f_{\text{max}}$, cycles remain effectively empty and orthogonality holds ($\Delta = 0$). The dimensionless coordination load is therefore

$$\rho(t) = \frac{f_{\text{PRO}}(t) + f_{\text{GOV}}(t)}{f_{\text{max}}} \quad (\text{A1})$$

2. Temporal independence and the geometric waiting distribution

The load above governs instantaneous rates, not waiting-time statistics. We model retry behavior as a memoryless process: each cycle presents pending events symmetrically, with service probability $1 - \rho$ per event. This is a modeling hypothesis about temporal independence, not a consequence of orthogonality. The waiting time follows a geometric distribution with mean

$$\langle k \rangle = \sum_{k=0}^{\infty} k \rho^k (1 - \rho) = \frac{\rho}{1 - \rho}. \quad (\text{A2})$$

3. The coordination cross-term

Each pending cycle incurs the fixed toll c_0 (the dimensionless Landauer scale; on thermal substrates this projects to $k_B T \ln 2$). The total waiting cost for both streams is

$$\Delta_{\text{total}}(t) = \frac{c_0 f_{\text{max}} \rho}{1 - \rho}. \quad (\text{A3})$$

The QSS baseline (equation (1)) already prices the linear coordination costs $Z_{\text{PRO}} + Z_{\text{GOV}} \approx c_0(f_{\text{PRO}} + f_{\text{GOV}})$ at $\rho \ll 1$. The cross-coupling $\Delta(t)$ is the excess beyond this baseline:

$$\Delta(t) = \Delta_{\text{total}}(t) - c_0 f_{\text{max}} \rho = \frac{c_0 f_{\text{max}} \rho^2}{1 - \rho}. \quad (\text{A4})$$

This recovers the three regimes of section VB: linear-response recovery as $\rho \rightarrow 0$, saturation divergence as $\rho \rightarrow 1$, and correlated-demand behavior under heavy-tailed waiting distributions.

4. Scope and validity

This construction assumes discrete cycles (the Margolus-Levitin bound) and symmetric, memoryless retry. Strongly correlated, heavy-tailed, or long-range-dependent arrivals fall outside its scope.